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Jnana Sangama, Belagavi, Karnataka - 590 018.



Project Work Report on

”IOT-Based Smart City management using ESP 32 Microcontroller ”

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

by

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CERTIFICATE

This is to certify that the project work entitled “**IOT-Based Smart City management using ESP 32 Microcontroller** ” is carried out by **Prateek S P(4JN22EC071), Rakesh R H (4JN22EC076), Varsha Vasanth (4JN22EC118), Varun Kumar (4JN22EC119)** ,towards the partial fulfillment of the requirement for the award of the Bachelor Degree in Electronics Communication Engineering as per the University regulations during May 2025. It is certified that all corrections / suggestions indicated for Internal Assessment have been approved as it satisfies the academic requirement in respect of Project Work.

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DECLARATION

We declare that this project report titled “Iot based Smart city Managment using ESP-32 Microcontroller” submitted in partial fulfillment of the degree of B.E. in Electronics and Communication Engineering is a record of original work carried out by me under the supervision of Prof. Sunil M D, and has not formed the basis for the award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, dueacknowledgements have been made wherever the findings of others have been cited.

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ABSTRACT

This project explores the integration of Internet of Things (IoT) technology into urban infrastructure to develop a Smart City system. This study focuses on three major urban challenges: waste management, parking congestion, and energy-efficient street lighting, by implementing IoT-based solutions using ESP32. By integrating smart sensors and real-time monitoring, this system enhances efficiency, reduces manual intervention, and promotes sustainable urban management. By implementing this system, citizens will experience cleaner surroundings, reduced traffic congestion, and improved safety, ultimately enhancing their quality of life. .

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PREAMBLE

1.1 Introduction

This project explores the integration of Internet of Things (IoT) technology into urban infrastructure to develop a Smart City system. This study focuses on three major urban challenges: waste management, parking congestion, and energy-efficient street lighting, by implementing IoT-based solutions using ESP32. By integrating smart sensors and real-time monitoring, this system enhances efficiency, reduces manual intervention, and promotes sustainable urban management. By implementing this system, citizens will experience cleaner surroundings, reduced traffic congestion, and improved safety, ultimately enhancing their quality of life.

1.2 Literature Survey

01. Gajula, Ramesh, Madhavi Karanam, Dinesh Chandra Vakkapatla, Kranthi Raj Endrapu, Vegesna Vijaya Rama Raju, Sadda Bharath Reddy, Saif Obayd Aljanabi, Sakshi Koli, and Shriya Mahajan. "Sustainable waste management solution: ESP32-powered smart bin." In *AIP Conference Proceedings*. AIP Publishing LLC, 2023.

In this system Ultrasonic are deployed for measurement of waste in dustbin and servo motor is used for opening and closing of bin. The servo motor effectively controlled the open in based on sensor input, ensuring hands-free operating and closing of the dustbin lid. Ultrasonic sensor not able to detect the type of waste. Improved Waste Management. Real time monitoring enables timely waste disposal chances of overflow and health hazards

02. Chaware, Sandeep M., Shriram Dighe, Akshay Joshi, Namrata Bajare, and Rohini Korke. "Smart garbage monitoring system using internet of things (IoT)." *International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering* 5, (2024)

Internet of Things (IoT) allows seamless integration of various systems and provides open access to data for digital service development. Designing a general IoT architecture is complex due to the diversity of devices, technologies, and services involved. Solid waste management is a major environmental and public health concern today. Traditional waste monitoring methods are manual, time-consuming, and inefficient. The proposed IoT-based garbage monitoring system automates waste tracking and sends real-time data to collection vehicles via a web page.

03.Simatupang, Joni Welman, and Aida Mahdalena Lubis. "IoT-based smart parking management system using esp32 microcontroller." In 2022 9th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI), pp. 305-310. IEEE, 2022.

The rising number of vehicles has led to wasted time, fuel, and increased air pollution due to drivers searching for parking spots. A smart, web-based parking management system was developed, allowing drivers to view and reserve parking slots without needing a mobile app. Ultrasonic sensors are used to detect slot availability, with a 100 cm threshold, and each slot has an indicator showing its status: vacant, reserved, or occupied. ESP32 modules control the sensors and indicators, and data is transmitted wirelessly using the MQTT protocol.

04.Parkash, Prabu V., and Dandu Rajendra. "Internet of things based intelligent street lighting system for smart city." International journal of innovative research in science, engineering and technology 5, no. 5 (2016).

Traditional systems keep lights ON from evening to morning, but lights are only needed when it's completely dark, leading to unnecessary power consumption. The proposed system uses IR transmitter and receiver pairs to detect the presence of vehicles or obstacles on the street. When movement is detected, the sensor sends data to a microcontroller which turns the street light ON, and automatically turns it OFF once the object is no longer present. The ON/OFF status of the lights can be monitored remotely in real-time via the internet, enabling smart control from anywhere, anytime.

05. Mehmood, Yasir, Farhan Ahmad, Ibrar Yaquob, Asma Adnane, Muhammad Imran, and Sghaier Guizani. "Internet-of-things-based smart cities: Recent advances and challenges." IEEE Communications Magazine 55, no. 9 (2017)

The Internet of Things (IoT) connects a wide range of digital devices with sensing, computing, and actuation capabilities to the Internet, enabling advanced services for smart cities. IoT services combined with big data analytics are driving global smart city initiatives by improving infrastructure, transportation, waste management, and overall quality of life. The article introduces a taxonomy to provide a general overview of the IoT paradigm in smart cities, covering integrated ICT, network types, opportunities, and key requirements. It reviews current efforts from standardization bodies and explores open-source IoT platforms used in smart city applications through real-world case studies. Finally, the article outlines major challenges and presents future directions for IoT-based smart cities.

1.3 Aim of the Project

Aim of the project is to develop an IoT-based Smart City system using ESP32 to improve waste management, parking efficiency, and street lighting for a smarter and more sustainable urban environment.

1.4 Objectives of the project

1. To develop a smart garbage system using an ultrasonic sensor and ESP32 to continuously monitor the fill level of dustbins.
2. To employ a password in a smart garbage system to open the dustbin using a servo motor.
3. To develop a smart parking system using IR sensors and a microcontroller to detect vehicle presence.
4. To develop an intelligent streetlight system using IR sensors and LDR.

1.5 Methodology

In this system Ultrasonic are deployed for measurement of waste in dustbin and servo motor is used for opening and closing of bin .System uses an ESP32 microcontroller to integrate sensors and communication modules for efficient waste management. An ultrasonic sensor measures the bin's fill level, while a moisture sensor differentiates between wet and dry waste.It monitors and manages parking spaces efficiently.Ultrasonic sensor used to detect vehicles in parking slot. Bluetooth is connected for real time communication between the system and the server.Uses ESP32 Microcontroller to control streetlights based on real time environmental data.It includes, LDR(Light dependent resistors) - To detect ambient light levels. PIR(Passive infrared sensor) - To detect vehicle movement.

1.6 Scope of Project/Applications

An IoT-based smart city project using ESP32 microcontroller has a wide scope and numerous applications, including waste management, smart car parking, and intelligent street lighting systems. By implementing sensors to monitor waste levels, guiding drivers

to available parking spots, and adjusting street light brightness based on ambient conditions, cities can become more efficient, sustainable, and livable. This project can improve public services, increase efficiency, reduce costs, and enhance citizen experience. It can also provide valuable data for informed decision-making, contributing to urban planning and development. The benefits of this project include reduced costs, improved safety, increased efficiency, environmental sustainability, and enhanced quality of life. Overall, this project has the potential to drive economic growth, foster innovation, and make cities more attractive to businesses, investments, and talent.

1.7 Limitations

The IoT-based smart city project faces several limitations, including potential security risks such as cyber attacks and data breaches, infrastructure challenges like reliable internet connectivity and power supply, and data management issues like handling large amounts of sensor-generated data. Additionally, scalability, interoperability, and power consumption are also significant concerns. The project requires regular maintenance and technical support, and high upfront costs may limit adoption. Furthermore, ensuring data privacy and complying with regulations are crucial. The risk of technological obsolescence also poses a challenge, as advancements in technology may render the system outdated, necessitating future upgrades or replacements. Addressing these limitations is essential to ensure the project's success and sustainability.

1.8 Organisation of the Report

This project presents the design and development of an IoT-based Smart City system using the ESP32 microcontroller to address three critical urban challenges: waste management, parking congestion, and energy-efficient street lighting. By integrating sensors such as ultrasonic, moisture, IR, LDR, and PIR with the ESP32, the system enables real-time monitoring, automation, and data transmission to cloud platforms or mobile applications. The smart waste management system ensures timely disposal and detects incorrect waste segregation, while the parking module provides real-time slot availability updates to reduce traffic and improve accessibility. Additionally, the intelligent street lighting system adjusts illumination based on

ambient light and vehicle movement, leading to significant energy savings. Overall, the project enhances urban sustainability, improves public services, and contributes to a cleaner, safer, and more efficient living environment.

Design and Implementation

2.1 Introduction

The rapid growth of urban populations has intensified the demand for smarter infrastructure and efficient public services. To address these challenges, this project focuses on the design and implementation of an IoT-based Smart City system using the ESP32 microcontroller. The system integrates various sensors and communication modules to automate and optimize three essential urban services: waste management, parking management, and street lighting. By utilizing real-time data collection and wireless communication, the system enables intelligent decision-making, reduces manual intervention, and promotes sustainability. The waste management module monitors dustbin levels and enforces proper waste segregation; the smart parking system detects vehicle presence and shares slot availability with users; and the intelligent street lighting system adapts illumination based on ambient light and traffic movement. The implementation of this system aims to improve urban living conditions by enhancing operational efficiency, reducing resource consumption, and providing a foundation for future smart city innovations.

2.2 System Design/Block Diagram

The block diagram illustrates an IoT-based Smart City system integrating three key subsystems: Garbage Level Monitoring, Waste Type Detection and Dustbin Lid Control, and Smart Parking and Streetlight Systems.

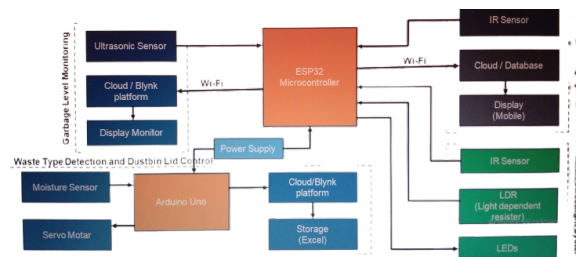


Figure 2.1: Block diagram

The Garbage Level Monitoring system utilizes an ultrasonic sensor connected to an ESP32 microcontroller to detect the fill level of garbage bins. This data is transmitted via Wi-Fi to a cloud or Blynk platform, which then displays it on a monitor or mobile device. The ESP32 plays a central role, handling data from multiple systems and ensuring wireless communication.

The Waste Type Detection and Dustbin Lid Control system uses a moisture sensor to determine the type of waste (wet or dry), and based on that, an Arduino Uno controls a servo motor to open or close the lid accordingly. This data is also sent to the cloud or Blynk platform and stored in an Excel sheet for record-keeping and analysis.

The Smart Parking System employs IR sensors to detect vehicle presence. These sensors communicate with the ESP32, which updates a **cloud database, and the data is shown on a mobile display. Meanwhile, the Smart Streetlight System includes IR sensors and LDRs (Light Dependent Resistors) to detect motion and ambient light, respectively. Based on these inputs, LED streetlights are controlled automatically for energy efficiency.

In summary, the diagram demonstrates an integrated IoT solution for managing urban infrastructure efficiently by connecting various sensors to microcontrollers (ESP32 and Arduino Uno), processing data in real-time, and displaying the results on cloud-connected platforms.

2.3 Circuit Description

In the garbage level monitoring section, an ultrasonic sensor is used to measure the fill level of dustbins. The data from this sensor is sent to the ESP32, which then transmits the information via Wi-Fi to a cloud platform like Blynk, enabling real-time monitoring. A display monitor shows the status of the dustbin for users or municipal staff. For waste type detection and dustbin lid control, an Arduino Uno is employed. It receives input from a moisture sensor to help identify the type of waste (e.g., wet or dry). Additionally, a servo motor is used to control the opening of the dustbin lid, which is likely password-protected. Data from the Arduino is processed and logged using a Python script and stored in Excel for future analysis. A power supply connects both controllers, ensuring continuous operation. The smart parking system uses IR sensors to detect whether a parking slot is occupied. These sensors communicate with the ESP32, which updates the parking status on a cloud database. The available or occupied status is then displayed on a mobile application or display panel for users to access in real time. Finally, the smart streetlight system integrates IR sensors and LDRs (Light Dependent Resistors). The LDR detects ambient light levels, while the IR sensor senses movement. Based on this input, LED streetlights are turned on or off to save energy. All these components communicate through the ESP32, forming a complete, connected smart city infrastructure.

HARDWARE REQUIREMENTS

1) ARDUINO UNO



Figure 2.2: Arduino UNO

The Arduino Uno is a microcontroller board based on the ATmega328P chip, which is an 8-bit microcontroller. It serves as the central control unit responsible for managing and coordinating the operations of various components connected to it. The board features 14 digital input/output pins and 6 analog input pins, allowing it to interface with a wide range of sensors and devices. It comes with 32KB of flash memory for storing programs, 2KB of SRAM for data processing, and 1KB of EEPROM for non-volatile data storage. Its simplicity and flexibility make it widely used in embedded systems and electronics projects.

2) ESP-32 Microcontroller

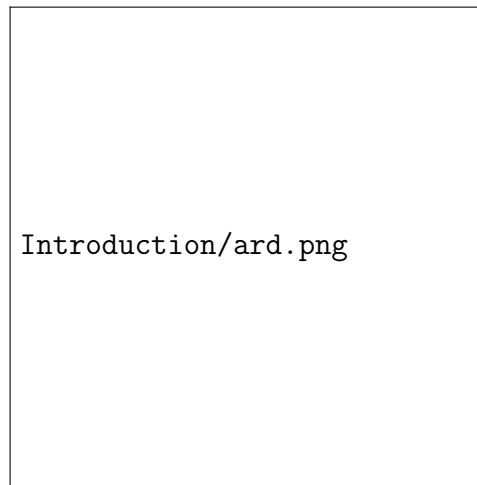


Figure 2.3: Arduino UNO

The ESP32 is a low-cost, low-power microcontroller with built-in Wi-Fi and Bluetooth, making it highly suitable for smart city applications such as smart parking and smart street lighting. In smart parking systems, the ESP32 collects data from sensors that detect vehicle presence and sends this information wirelessly to a central server or mobile app. This helps drivers find available parking spaces easily, reducing traffic congestion and fuel consumption. In smart street lighting, the ESP32 controls lights based on environmental conditions such as light intensity or motion detection. It enables automatic switching and dimming of street lights, which saves energy and reduces electricity costs. The wireless connectivity of the ESP32 allows remote monitoring and control of both systems. Overall, the ESP32 improves efficiency, automation, and sustainability in smart parking and street lighting systems.

3) IR sernsors



Figure 2.4: Arduino UNO

IR (Infrared) sensors are widely used in smart parking and smart street lighting systems for detection and automation purposes. In smart parking, IR sensors detect the presence or absence of vehicles in parking slots by sensing reflected infrared radiation. This information is sent to a controller, helping identify available parking spaces in real time. It reduces manual monitoring and improves parking management efficiency. In smart street lighting, IR sensors are used to detect motion of vehicles or pedestrians. When movement is detected, the lights are automatically turned ON or brightened, and they switch OFF or dim when no motion is present. This helps in saving energy and increasing the lifespan of street lights. Thus, IR sensors play a key role in improving automation, safety, and energy efficiency in smart city applications.

4) LEDs



Figure 2.5: Arduino UNO

LEDs are an important component of smart street light systems due to their high ef-

efficiency and long lifespan. In street lighting, LEDs provide bright illumination while consuming significantly less power compared to traditional lamps. They can be easily controlled using microcontrollers to turn ON, OFF, or dim based on time, motion, or ambient light conditions. This helps in reducing energy consumption and operational costs. LEDs also generate less heat, which increases safety and durability. Their fast response time makes them suitable for motion-based lighting systems. Additionally, LEDs require less maintenance because of their longer working life. Overall, LEDs enhance energy efficiency, reliability, and sustainability in smart street light systems.

5) Ultrasonic Sensor

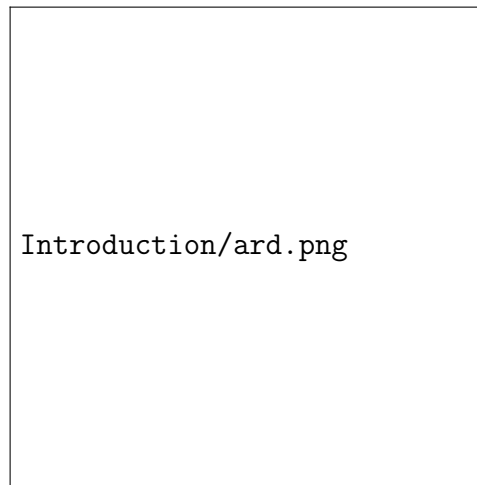


Figure 2.6: Arduino UNO

Ultrasonic sensors are commonly used in smart dustbin systems to measure the fill level accurately. They work by emitting ultrasonic sound waves and measuring the time taken for the echo to return after hitting the waste surface. Based on this time, the distance between the sensor and the garbage level is calculated. This data helps determine how full the dustbin is in real time. The information can be sent to a central monitoring system for timely waste collection. It reduces unnecessary collection trips and improves waste management efficiency. Ultrasonic sensors are reliable and work well in different lighting conditions. Overall, they help maintain cleanliness and support efficient smart city waste management systems.

6) Servo Motor

A servo motor is used in a smart dustbin system to automatically open and close the lid of the bin. It operates by rotating to a specific angle when it receives a control signal from a microcontroller. In this system, the servo motor opens the lid when a person approaches the dustbin or when a sensor detects motion. This allows users to dispose of

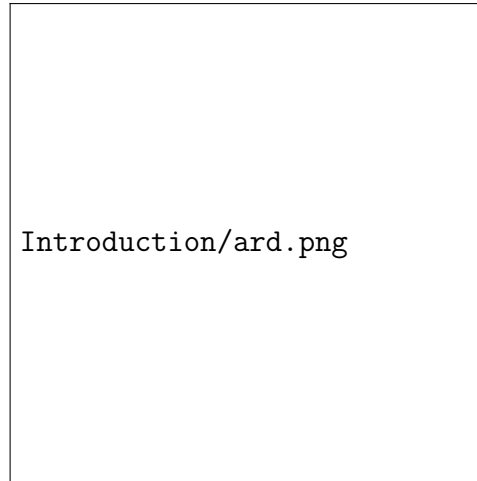


Figure 2.7: Arduino UNO

waste without touching the bin, improving hygiene. After a short time, the servo motor closes the lid automatically. Servo motors are precise, reliable, and consume less power. They are easy to interface with controllers like ESP32 or Arduino. Thus, servo motors enhance automation, cleanliness, and user convenience in smart dustbin systems.

2.4 Implementation

The implementation of this IoT-based Smart City project involves the integration of various sensors with the ESP32 microcontroller to automate and optimize urban services. For waste management, ultrasonic sensors monitor dustbin fill levels, while a soil moisture sensor differentiates between wet and dry waste; a servo motor operates the bin lid through password authentication, and any improper disposal is logged and fined automatically. In the smart parking system, IR sensors detect vehicle presence in each slot, with real-time availability updates sent to a mobile app via ESP32 and Wi-Fi, helping reduce traffic congestion. The intelligent street lighting system uses LDR sensors to detect ambient light and PIR sensors to identify motion, allowing the ESP32 to control street-light brightness dynamically, turning lights on only when needed to conserve energy. All modules communicate wirelessly, enabling real-time data transmission, centralized monitoring, and efficient urban service delivery.

2.5 summary

The design and implementation of this IoT-based Smart City project using the ESP32 microcontroller focuses on creating an integrated system to address key urban challenges: waste management, parking, and street lighting. The system is designed using ultrasonic, IR, LDR, PIR, and moisture sensors to collect real-time data, which is processed by the ESP32 and transmitted wirelessly to cloud platforms and mobile applications. In waste management, the system monitors bin levels, detects improper disposal, and automates access control. The smart parking module detects vehicle presence and provides real-time slot availability to users, while the intelligent street lighting system adjusts brightness based on ambient light and motion detection to save energy. The implementation ensures efficient resource usage, reduced manual intervention, and enhanced public services, contributing to a smarter, safer, and more sustainable urban environment.

Results and Discussions



Figure 3.1: Prototype of our Project

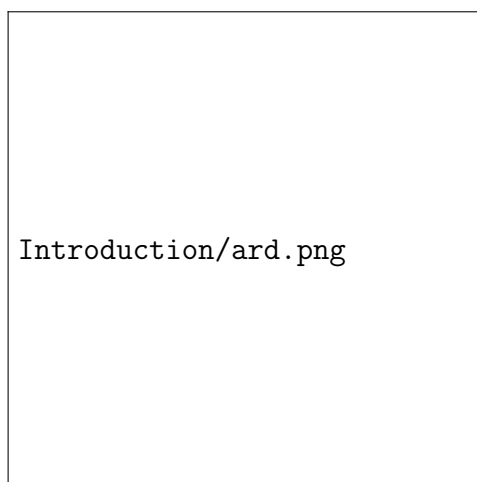


Figure 3.2: Result of First Objective

The first objective of the project, which focuses on garbage level monitoring using an ultrasonic sensor interfaced with an ESP32 microcontroller, was successfully implemented. The system transmitted data via Wi-Fi to the Blynk platform, allowing real-time monitoring of the dustbin's fill level. The recorded result showed a bin level of 48 per- cent, demonstrating accurate sensing and data transmission. This enables timely garbage collection and helps prevent overflow.



Figure 3.3: Smart bin



Figure 3.4: Result of second Objective

For the second objective, a more advanced smart garbage system was proposed. It includes features such as password-protected access using a servo motor, detection of wet waste with a soil sensor, and automatic logging of violations in an Excel sheet. These enhancements aim to promote responsible waste disposal and enable efficient monitoring by authorities.

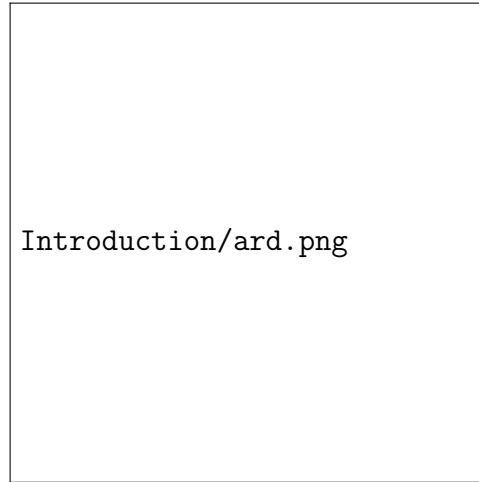


Figure 3.5: Result of 3rd Objective

The result of the third objective demonstrates the successful monitoring and display of the availability status of three parking slots in the smart parking system. Each slot's status is clearly shown as either available or not available based on real-time sensor data. The system accurately records the in-time when a vehicle enters a parking slot and the out-time when the vehicle leaves. This information is updated continuously and displayed on the monitoring interface. The real-time tracking helps users quickly identify free parking spaces. It also improves parking management and reduces waiting time for drivers. The recorded time data can be used for billing and analysis purposes. Overall, the result confirms that the system effectively manages and displays parking slot availability with accurate time logging.

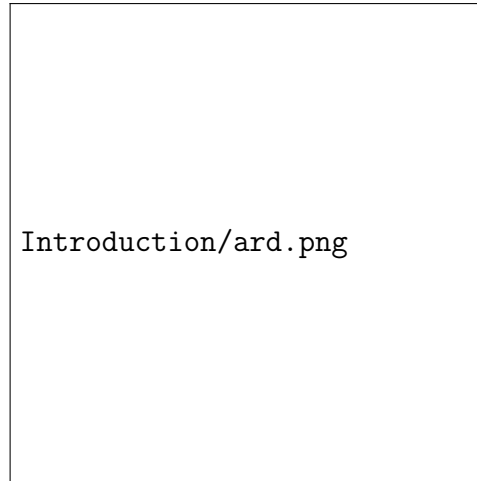


Figure 3.6: Result of 4th Objective

The result of the fourth objective demonstrates the effective working of the smart street light system using object detection and LDR sensing. The system turns the street light ON only during night conditions, which is detected using the LDR sensor. When an object such as a vehicle or pedestrian is detected, the light automatically switches ON. If no object is present, the light remains OFF, helping to save energy. During daytime, the street light stays OFF regardless of object detection. This automatic operation reduces power consumption and human intervention. The system responds accurately to changes in light intensity and object presence. Overall, the result confirms that the smart street light system operates efficiently and supports energy-saving automation.

Conclusion

The implementation of an IoT-based smart city project encompassing waste management, smart parking systems, and intelligent street lighting represents a transformative step toward urban sustainability, efficiency, and improved quality of life for citizens. This integrated approach leverages Internet of Things (IoT) technologies to address critical urban challenges through real-time data collection, analysis, and automated responses. In the waste management component, smart bins equipped with sensors monitor fill levels and communicate with a centralized system to optimize collection routes, reducing fuel consumption, operational costs, and carbon emissions while preventing overflow and promoting a cleaner environment. The smart parking system utilizes sensors and mobile applications to identify available parking spaces and guide drivers accordingly, significantly reducing traffic congestion, fuel wastage, and the time spent searching for parking spots. This not only enhances convenience for drivers but also contributes to improved air quality and more organized urban mobility. Meanwhile, the smart street lighting system employs motion sensors, ambient light detectors, and remote control mechanisms to regulate lighting based on real-time environmental and human activity data. By dimming lights during low-traffic periods and increasing brightness when necessary, cities can achieve substantial energy savings, lower maintenance costs, and extend the lifespan of lighting infrastructure without compromising public safety.

4.1 Reference

1. Gajula, Ramesh, Madhavi Karanam, Dinesh Chandra Vakkapatla, Kranthi Raj Endrapu, Vegesna Vijaya Rama Raju, Sadda Bharath Reddy, Saif Obayd Aljanabi, Sakshi Koli, and Shriya Mahajan. "Sustainable waste management solution: ESP32-powered smart bin." In AIP Conference Proceedings. AIP Publishing LLC, 2023.
2. Chaware, Sandeep M., Shriram Dighe, Akshay Joshi, Namrata Bajare, and Rohini Korke. "Smart garbage monitoring system using internet of things (IoT)." International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering 5, (2024).
3. Simatupang, Joni Welman, and Aida Mahdalena Lubis. "IoT-based smart parking management system using esp32 microcontroller." In 2022 9th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI), pp. 305-310. IEEE, 2022.
4. Parkash, Prabu V., and Dandu Rajendra. "Internet of things based intelligent street lighting system for smart city." International journal of innovative research in science, engineering and technology 5, no. 5 (2016).
5. Mehmood, Yasir, Farhan Ahmad, Ibrar Yaquob, Asma Adnane, Muhammad Imran, and Sghaier Guizani. "Internet-of-things-based smart cities: Recent advances and challenges." IEEE Communications Magazine 55, no. 9 (2017).